

TIC 59603820

PERIOD ALIAS

Eclipsing binary candidate dossier - Star Vetter

BLS EPHEMERIS

Period 7.689930 d (ALIASSED, see below)
-> corrected period: 15.379861 d
Epoch (t0) 1517.5356 BTJD
Duration 0.1200 d
Depth 17.353 % (173,531 ppm)
SNR 547.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

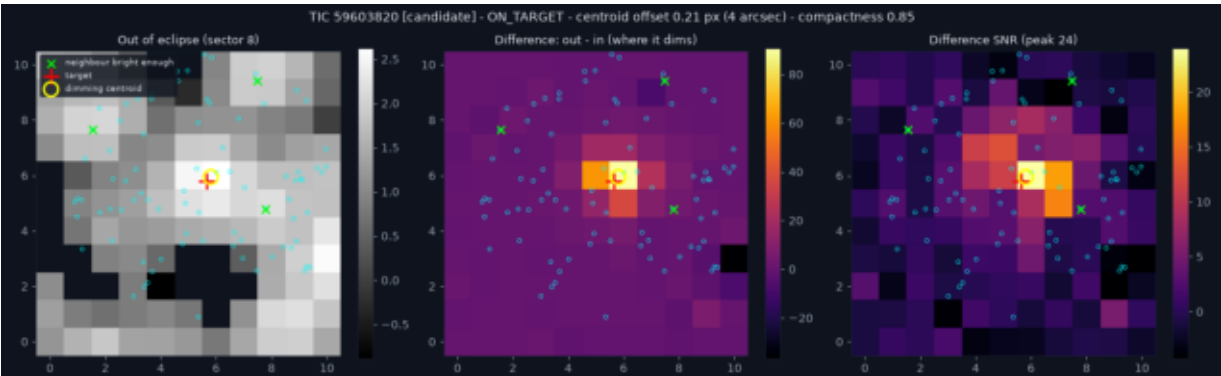
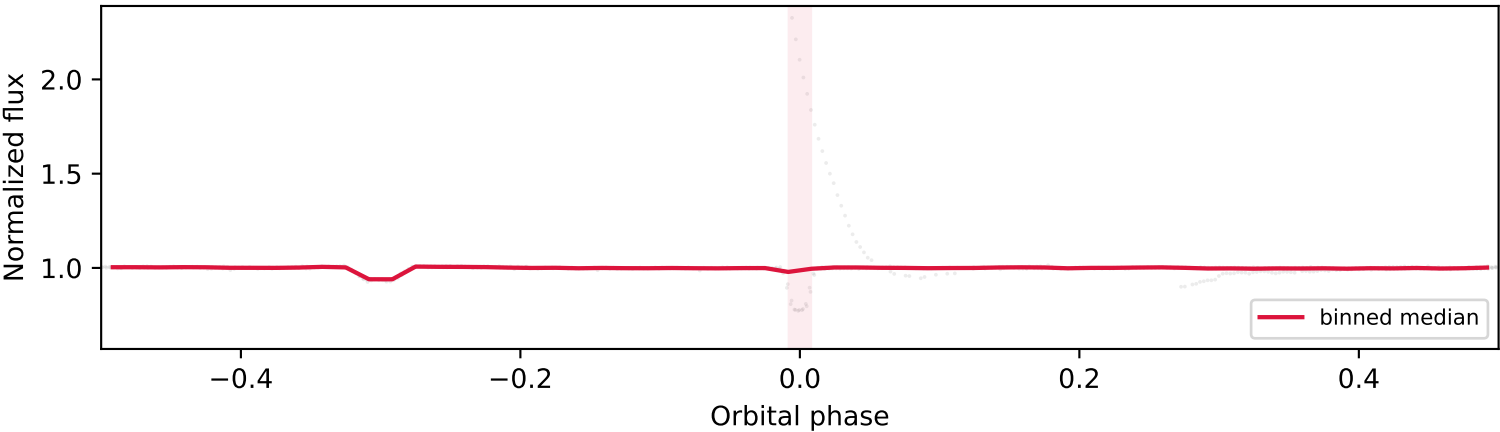
Eclipses observed 3 (gate >= 3)
Robust depth ratio 1.10 (gate >= 0.50)
Robust depth sigma 150.1 (gate >= 7.0)
Odd/even z-score 48.39 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.29, 6.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 8 light-curve dip in this sector: 111.5 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 23.99 (gate >= 5.0) Compactness 0.85 (gate >= 0.50)
Dimming centroid offset 0.21 px (4 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 59603839
2.29 px away, dTmag +1.40 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 48.39 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 15.379861 d, in any write-up.